ECON 625: Mathematical Economics

Old Dominion University • Strome College of Business
Fall 2026 • First 8-Week Session • Asynchronous Online

Course	ECON 625, 3 credit hours, CRN 20705
Term dates	Monday, August 24 – Friday, October 16, 2026 Final exam window: Friday, October 9, 11:59 p.m. – Friday, October 16, 11:59 p.m. ET
Modality	Fully asynchronous online via Canvas. No scheduled meeting times.
Instructor	Dr. Rex W. Sitti, Assistant Professor of Economics
Email	rsitti@odu.edu — the fastest way to reach me
Response time	Within 24 hours on weekdays; within 48 hours on weekends
Office hours	Virtual, by appointment via Zoom. Email me to arrange a time, then join at: odu.zoom.us/j/929 7829 9777 (Meeting ID 929 7829 9777; passcode embedded in the link). Standing drop-in hours: **Tuesdays, 2:00–3:30 p.m. ET**, same link, no appointment needed.
Feedback turnaround	Problem sets graded within 5 days of the due date; exams within 7 days

1 Course Description

This graduate course builds a rigorous foundation in the mathematical methods used in advanced economic analysis. You will learn to formulate economic problems mathematically, solve them, and interpret the results in economic terms. Topics include equilibrium analysis, matrix algebra, comparative statics, unconstrained and constrained optimization, differential and difference equations, and an introduction to optimal control theory. The emphasis throughout is on economic interpretation: a correct derivative that you cannot explain economically is only half an answer.

A note on the compressed format

This is a 15-week course delivered in 8 weeks. Plan on **15–18 hours per week** of reading, practice problems, and assignments. Four of the seven modules cover three textbook chapters each. If you are working full time or taking other compressed courses this session, please look carefully at the schedule in Section 12 before committing to the course.

2 Prerequisites

Graduate standing in economics or a related field, or permission of the instructor. I assume you are comfortable with:

- single- and multivariable differential calculus (chain rule, partial derivatives, total differentials);
- basic integration techniques;
- intermediate microeconomic theory (utility maximization, cost minimization, market equilibrium).

A self-check diagnostic is posted in Module 0. It is ungraded, takes about 45 minutes, and is the single best predictor of how the first three weeks will go. Take it before the add/drop deadline on **Friday, August 28**.

3 Required Materials

Textbook

Chiang, Alpha C., and Kevin Wainwright (2005). *Fundamental Methods of Mathematical Economics*, 4th Edition. McGraw-Hill Education.
ISBN-13: 978-0-07-010910-0 ISBN-10: 0-07-010910-9
Publisher page. Any copy of the 4th edition is fine, including used or rental. Check that the edition matches — chapter numbering in the 3rd edition differs, and every reading in this syllabus is keyed to the 4th.

Supplemental (optional)

Simon, Carl P., and Lawrence Blume (1994). *Mathematics for Economists*. W.W. Norton. Useful if you want more proof-level detail, especially for Modules 5 and 7.

Technology

- **Canvas** — all content, submissions, and grades. Check announcements at least every other day.
- **Zoom** — office hours.
- **A spreadsheet program** (Excel or Google Sheets) — for numerical exercises in Modules 2, 6, and 7.
- **A way to produce legible mathematics** — LaTeX, MathType, or neatly handwritten work scanned to PDF. See Section 5.
- **A reliable internet connection** — exams are timed and the clock does not stop for connection problems (see Section 7).

4 Learning Outcomes

By the end of this course you will be able to:

LO1. **Translate** a verbal economic problem into a formal mathematical model and state its assumptions.

LO2. **Solve** systems of linear economic equations using matrix inversion, Cramer's rule, and input-output methods, and evaluate whether a solution exists and is unique.

LO3. **Derive** comparative-static results for general-function models using total differentiation and the implicit function theorem, and **sign** those results.

LO4. **Solve** unconstrained and equality-constrained optimization problems and **verify** second-order conditions.

LO5. **Apply** the envelope theorem and duality results (Roy's identity, Shephard's lemma) to consumer and producer problems.

LO6. **Analyze** dynamic economic models in continuous and discrete time and **assess** the stability of their steady states.

LO7. **Set up** a simple intertemporal optimization problem using the maximum principle and interpret the costate variable economically.

LO8. **Communicate** mathematical results in clear economic language, with reasoning a reader can follow.

Where each outcome is assessed

Outcome	Module	Assessed by
LO1	1	Problem Set 1
LO2	2	Problem Set 2; Midterm
LO3	3	Problem Set 3 (including the written analysis); Midterm
LO4	4, 5	Midterm; Problem Set 4
LO5	5	Problem Set 4 (proof questions); Final
LO6	6, 7	Problem Set 5; Final
LO7	7	Final
LO8	3, 5, 7	Written analysis (PS3), proof questions (PS4), interpretation questions on both exams

5 How the Course Runs

Weekly rhythm

Every module follows the same pattern, so you always know what is due when:

- **Monday:** module opens; read the assigned chapters and watch the posted worked-example videos.
- **Midweek:** attempt the problem set early enough that you can bring questions to Tuesday drop-in hours or email me with time to spare.
- **Sunday, 11:59 p.m. ET:** problem set or exam due.

All deadlines are **Eastern Time**. If you are in another time zone, set Canvas to display ET and work backward from there.

Submission standards

- Submit **one PDF per assignment** through Canvas. No photos pasted into Word, no `.heic` files, no multi-file uploads.
- Handwritten work is welcome if it is legible and scanned right-side-up. Use a scanning app, not a raw camera photo.
- **Show your work.** Partial credit is awarded for reasoning, and an unsupported final answer earns close to nothing even when it is correct.
- Label problems clearly and in order. Circle or box final answers.
- For every comparative-static or optimization result, include one sentence stating what it means economically. This is part of the grade, not a courtesy.

6 Module Schedule

Module 0 — Start Here (open all term)

Course orientation, a short syllabus quiz, and the prerequisite self-check diagnostic. Both are ungraded, but the syllabus quiz must be completed before Module 1 unlocks. Finish Module 0 in the first two days of the session.

Module 1 — Foundations of Mathematical Economics Week 1

Chiang & Wainwright, Ch. 1–3. The mathematical method in economics, model building and abstraction, partial versus general equilibrium, static systems.
Due: Problem Set 1.

Module 2 — Linear Models and Matrix Algebra Week 2

Ch. 4–5. Systems of linear equations, determinants, matrix inversion, Cramer's rule, Leontief input-output analysis, existence and uniqueness of solutions.
Due: Problem Set 2.

Module 3 — Comparative Statics and Differentiation Week 3

Ch. 6–8. Derivatives and rules of differentiation, total differentials, the implicit function theorem, elasticities, comparative statics of general-function models.
Due: Problem Set 3, which includes a short written analysis (roughly one page) interpreting one comparative-static result economically.
This is the heaviest reading week of the term. Start Monday.

Module 4 — Unconstrained Optimization Week 4

Ch. 9–11. First- and second-order conditions, exponential and logarithmic functions, Cobb-Douglas forms, optimization with several choice variables, cross-partial interpretation.
Due: Midterm Exam (cumulative, covering Modules 1–4).
There is no problem set this week so that you can study. Practice problems with full solutions are posted instead.

Module 5 — Constrained Optimization and Duality Week 5

Ch. 12–13. Lagrange multipliers and their interpretation, bordered Hessians, comparative statics of optimal solutions, the envelope theorem, duality, Roy's identity, Shephard's lemma.
Due: Problem Set 4, which includes two short proof questions.

Module 6 — Dynamics I: Continuous Time Week 6

Ch. 14–16. Integration and economic applications, first-order and higher-order differential equations, steady states, phase diagrams, stability analysis, growth models.
Due: Problem Set 5 (applied dynamic analysis).

Module 7 — Dynamics II: Discrete Time and Optimal Control Week 7 (Oct. 5–9)

Ch. 17–18. First-order and higher-order difference equations, the cobweb model, stability conditions in discrete time.
Ch. 20. Introduction to optimal control: the Hamiltonian, Pontryagin's maximum principle, transversality conditions, the costate variable as a shadow price.
Chapter 19 (simultaneous differential and difference equations) is not covered in this course. It is not assigned and it is not tested. If you go on to work with dynamic systems, it is the natural next chapter to read on your own.
Due: All Module 7 material is posted Monday, October 5 and instruction closes Friday, October 9. An ungraded practice set with full solutions is posted the same Monday so you can self-check before the final exam window opens.
Week 8 (Oct. 12–16) contains Fall Break on Monday and Tuesday and is reserved for the final exam. There is no new material that week.

7 Assessment and Grading

Component	Points	Weight
Problem Sets 1–5 (80 points each; all five count)	400	40%
Midterm Exam (end of Module 4; cumulative, covering Modules 1–4)	300	30%
Final Exam (Week 8; cumulative, covering Modules 1–7)	300	30%
Total	**1,000**	**100%**

Problem sets

Five sets, one per module, due in Weeks 1, 2, 3, 5, and 6. Each is worth 80 points and **all five count**, for 400 points total. There is no dropped score: each set covers material that appears nowhere else, and a missed set is material you will meet again on the final. The written analysis in PS3 and the proof questions in PS4 are components of those sets, not separate assignments.

Exams

- **Format:** open-book and open-notes, closed-collaboration. You may use the textbook, your own notes, and a calculator. You may not use another person, a solutions manual, or a generative AI tool.
- **Coverage:** both exams are cumulative. The **Midterm** covers Modules 1–4. The **Final** covers Modules 1–7, that is, everything in the course. Material from the first half is not retired after the midterm; it is the foundation the second half is built on.
- **Midterm window:** opens Thursday, September 17, 12:01 a.m. and closes Sunday, September 20, 11:59 p.m. ET. To get the full three hours you must start by 8:59 p.m. on September 20.
- **Final exam window:** opens Friday, October 9, 11:59 p.m. and closes Friday, October 16, 11:59 p.m. ET. The window deliberately spans Fall Break so you can choose your own sitting.
- **Duration:** once you begin, you have **3 hours** to finish. The timer does not stop for any reason, and it does not stop at the closing time either — to get the full three hours on the final you must **start by 8:59 p.m. on October 16**. Do not start an exam you do not have three uninterrupted hours to complete.

- **Work shown:** both exams require handwritten derivations uploaded as a single PDF within 15 minutes of submitting the Canvas portion. Practice this workflow during the Module 0 syllabus quiz so it is not new to you on exam day.
- **Formula sheet:** a formula sheet I provide is posted in Module 4 and attached to both exams. You do not need to memorize functional forms.

Grading scale

A: 93–100	A–: 90–92	B+: 87–89	B: 83–86
B–: 80–82	C+: 77–79	C: 73–76	C–: 70–72
F: below 70			

Final grades are not rounded beyond the scale above and are not curved. There is no extra credit.

Graduate grading standards

Two university rules matter more in graduate school than the letter on any single assignment:

- **Grades of D+ or lower are not assigned to graduate students.** The scale above therefore runs from A to C–, and anything below 70 is recorded as an F. See the ODU Grade Key.
- **A cumulative 3.00 GPA is required** to remain in good academic standing and to apply for graduation. A grade in the C range is passing, but it pulls against that requirement, and your program may require you to retake a course in which you earn one. If you are heading toward the C range at the midterm, come talk to me while there is still time to change the outcome.

Program-specific requirements are set by your department, not by this course; check your graduate program handbook for how a low grade in a required course is handled.

8 Course Policies

Late work

Problem sets are accepted up to 48 hours late with a 20% deduction. After 48 hours the Canvas submission closes and the score is zero. Because no problem set is dropped, a missed set is costly: if something is going wrong, email me **before** the deadline rather than after it. I would far rather grant you an extension on Friday than write off a zero on Wednesday. Exams cannot be submitted late. If a documented emergency (illness, hospitalization, family emergency, military obligation) prevents you from taking an exam in its window, contact me **before the window closes** if at all possible, and we will arrange a make-up.

Regrade requests

If you believe a grading error was made, email me within **seven days** of the grade being posted, identifying the specific problem and explaining the mathematical reasoning you believe was misread. I will regrade the entire submission, which can move the score in either direction.

Academic integrity

You are bound by the ODU Honor Pledge. On this course specifically:

- **Problem sets:** you may discuss approaches with classmates. You must write up every solution independently and in your own hand. Copied write-ups — including identical algebra errors — are treated as unauthorized collaboration.
- **Solutions manuals and answer-key websites** (Chegg, Course Hero, Numerade, and similar) are prohibited for all graded work, including uploading questions to them.
- **Exams:** entirely independent work.

Generative AI

You may use generative AI (ChatGPT, Claude, Gemini, and similar) as a tutor on problem sets: to explain a concept, check an algebra step, or suggest a different way to see a proof. You may not submit AI-generated work as your own. Concretely:

- **Permitted:** "Explain why the bordered Hessian condition differs from the unconstrained case."
- **Not permitted:** pasting a problem set question and submitting the output, in whole or in part.
- **Required disclosure:** if you used AI on an assignment, add one line at the end naming the tool and what you used it for. Disclosure is not penalized; undisclosed use is an integrity violation.
- **Exams:** no AI use of any kind.

Two practical warnings. These tools are confidently wrong on second-order conditions, sign restrictions, and transversality conditions — exactly the material that separates a B from an A here. And the exams are where you find out whether you learned the material or watched something else do it.

Communication and netiquette

Email is the channel for this course. Use your ODU email or the Canvas Inbox and put "ECON 625" in the subject line. When a question comes up more than once, I post the answer as a Canvas announcement so the whole class has it — so check announcements before you write, and expect to see your question answered there if it turns out to be a common one. For anything that needs a derivation walked through rather than typed out, come to Tuesday drop-in hours; ten minutes on Zoom beats a long email thread.

Recordings

Office hours are not recorded unless all participants agree in advance. Posted lecture and worked-example videos are for enrolled students only and may not be redistributed.

9 University Policies

Academic integrity

All students are expected to uphold the Old Dominion University Honor Code. Violations, including cheating, plagiarism, and unauthorized collaboration, are handled under university policy: odu.edu/academic-integrity.

Disability services

ODU provides reasonable accommodations to students with documented disabilities. Register with the Office of Educational Accessibility (OEA) and send me your accommodation letter as early in the session as possible — in an 8-week course, an accommodation letter that arrives in Week 6 cannot be applied retroactively. odu.edu/accessibility.

Title IX and non-discrimination

ODU does not discriminate on the basis of race, color, national origin, sex, disability, age, religion, or veteran status. Please be aware that faculty are required to report disclosures of sexual harassment or sexual violence to the Title IX Coordinator. If you would prefer to speak confidentially, University Counseling Services and the Women's Center are confidential resources. odu.edu/titleix.

Religious observance

If a deadline conflicts with a religious observance, notify me within the first week of the session and we will arrange an alternative date.

Important dates

These deadlines are for the **First 8-Week Session** and differ from the full-term calendar. Do not go by the 16-week dates.

Date	Deadline
Monday, August 24	Term begins; first day of class
Friday, August 28	Add/drop deadline. No "W" on transcript, 100% tuition refund
Saturday, August 29	Withdrawal period opens. A "W" now appears on the transcript
Monday, September 7	Labor Day, university holiday. Because this course is asynchronous, no deadline shifts
Friday, October 2	**Withdrawal deadline.** Last day to withdraw with a "W"
Saturday, October 10 – Tuesday, October 13	Fall Break, no classes. The final exam window stays open throughout
Friday, October 16	Last day of the session; final exam closes at 11:59 p.m. ET
Tuesday, October 20	Grades due to the Registrar

Note the withdrawal deadline of October 2. It falls at the end of Week 6, which is deliberate on the university's part and useful on yours: midterm grades are returned by roughly September 27, so you will know exactly where you stand with several days left to decide. If the midterm goes badly, email me before October 2 rather than after it.

The academic calendar publishes a 50% tuition refund deadline for the full term but not for the 8-week sessions. If you are considering withdrawing and the refund matters, confirm the date with the Office of the University Registrar or Student Accounts before you act. There is no separate exam period for this session — the final is due by the last day of class.

Inclement weather and emergencies

Because this course is asynchronous, university closures do not change deadlines unless I announce otherwise in Canvas. If a widespread outage affects Canvas itself, I will extend the affected deadline and post an announcement.

10 Support Resources

- University Counseling Services: odu.edu/counselingservices
- Writing Center: odu.edu/writing-center
- University Libraries: odu.edu/library
- IT Help Desk: odu.edu/ts/help-desk — contact them first for Canvas or proctoring problems, and forward me the ticket number
- Graduate School resources: odu.edu/graduateschool/graduate-student-resources

11 Tips for Success

1. **Do not fall behind.** This material is strictly cumulative and the session is half-length. A week lost in Module 3 is very hard to recover by Module 5.
2. **Work problems with the book closed.** Reading a worked solution and following it is not the same skill as producing it. Attempt each problem cold first.
3. **Work the odd-numbered exercises** beyond the assigned set. Answers are in the back of Chiang.
4. **Interpret every result.** After each derivation, write one sentence explaining what it means economically. This is what exams reward.
5. **Ask early.** A five-minute Zoom call in Week 2 saves five hours in Week 5.
6. **Form a study group.** Discussing problems is permitted and effective. Writing up solutions together is not.

12 Tentative Schedule

Week	Mod.	Topics (Chiang & Wainwright)	Due 11:59 p.m. ET
1 Aug. 24–30	0 & 1	Orientation; mathematical method, economic models, equilibrium analysis (Ch. 1–3)	Syllabus quiz; PS 1 *Sun. Aug. 30*
2 Aug. 31–Sep. 6	2	Linear models, matrix algebra, input-output analysis (Ch. 4–5)	PS 2 *Sun. Sep. 6*
3 Sep. 7–13	3	Comparative statics, differentiation, general-function models (Ch. 6–8). *Labor Day Sep. 7 — no deadline change*	PS 3 + written analysis *Sun. Sep. 13*
4 Sep. 14–20	4	Unconstrained optimization, exponential and logarithmic functions (Ch. 9–11)	**Midterm**, Modules 1–4 *window Sep. 17–20*
5 Sep. 21–27	5	Constrained optimization, envelope theorem, duality (Ch. 12–13)	PS 4 + proof questions *Sun. Sep. 27*
6 Sep. 28–Oct. 4	6	Integral calculus, differential equations, phase diagrams, stability (Ch. 14–16)	PS 5 *Sun. Oct. 4*
7 Oct. 5–9	7	Difference equations and the cobweb model (Ch. 17–18); optimal control (Ch. 20). *Instruction closes Fri. Oct. 9*	Practice set (ungraded)
8 Oct. 12–16	—	Fall Break Oct. 10–13; review and exam. No new material	**Final Exam**, Modules 1–7 *window Oct. 9–16*

This syllabus is subject to change at the instructor's discretion. Any change will be announced in Canvas and a revised syllabus posted. Deadlines will only ever move later, never earlier.